Group Q: Phase 2 Report

Eco-log

Author revision copy – not for submission

Group Q

Course: 5ARF0 Ethics of AI: Theory and Practice
Eindhoven University of Technology

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Name	Student ID	Email
Yehao Chen	2216655	y.chen13@student.tue.nl
Dennis de Hoon	1732773	d.b.a.d.hoon@student.tue.nl
Gerald Tobenna Eluagu	2419289	g.t.eluagu@student.tue.nl
Peng Wang	2342944	p.wang1@student.tue.nl

Team revision copy: Sections B and C have been aligned with the two-tree research-discovery design. Section C now includes deployment, cold start, replication choices, governance and evaluation. Dennis's Section D remains for his final revision. Gerald coordinates the other components; Section F incorporates the latest five-step Eco-log narrative; G still needs matching screenshots, and H requires contributor records. The editorial checklist at the end is for team review and must be removed before submission.

Word counts use TeXcount narrative-text counts, excluding headings, citations, references, captions and editorial notes. Recount after further edits.

A. Project Title

Eco-log: Finding Related TU/e Research and Reusable Resources

B. Public Summary

Word count: 137/ 150

Our Environmental Sustainability project introduces Eco-log, an AI-supported web tool that helps TU/e researchers connect planned studies with existing work. It addresses a practical question: how can researchers discover opportunities to reuse data and laboratory equipment before committing additional resources? Two interactive trees show similarities and possible differences relative to the sources found. Researchers can explore relevant studies and potential collaborators, while retaining the choice to conduct replication.

For the TU/e community, the intended benefit is to turn sustainability goals into research-planning decisions. Identifying reusable resources could reduce additional material and energy use, but savings would need to be measured. The main challenges are keeping records updated, protecting unpublished ideas, explaining incomplete coverage and preventing overreliance on automated suggestions. The tool supports researcher judgment; it cannot establish originality or prove an environmental benefit by itself.

C. Comprehensive Summary

Word count: 1670/ 2000

Problem Definition and Objective(s)

A student planning a study may find publications on the same topic while overlooking a useful method, dataset or piece of equipment in another TU/e department. The TU/e research portal already contains publications, projects, equipment and facilities (TU/e, n.d.-b); the problem we address is discovering relevant connections before the student knows the right author or search term. Its extent at TU/e still requires local investigation. We therefore propose a design strategy for research discovery rather than claim that researchers routinely waste resources.

Eco-log would let students and researchers describe a planned study and explore related work through a Similarity & Reuse Tree and a Distinctiveness & Research Gaps Tree. The intended change is to make potential reuse visible while research plans remain open to revision. A low-temperature cathode study, for example, might benefit from an experimental setup or dataset used in earlier battery research. Researchers would still need to check whether the method transfers and whether the associated resources are available.

Students gain a starting point for discussion with supervisors; researchers can discover potential collaborators and resources. Librarians would help with discovery and metadata, equipment managers would confirm access conditions, and data stewards would advise on sharing. Their participation is necessary: a matching record cannot establish permission or suitability for reuse. The tool would support planning, without assessing eligibility for funding or deciding whether a project deserves to proceed.

Background of Current Approaches and Knowledge

The research portal provides an institutional starting point, while OpenAlex could supplement the published literature beyond TU/e (OpenAlex, n.d.). These sources serve different purposes: wider literature contextualises a research question; institutional records help locate local expertise and resources. Neither establishes complete coverage of ongoing or confidential work. Pure and Research Cockpit should also be distinguished. Pure offers research-information services; TU/e's Research Cockpit supports processes including data management and ethical applications (TU/e, n.d.-a, 2026b). A link from an existing workflow would not imply access to its underlying data.

SPECTER provides a research basis for representing scientific documents through citation-informed learning (Cohan et al., 2020). SciRepEval evaluates representations across different scientific tasks (Singh et al., 2023). Neither establishes the accuracy of matching short TU/e project descriptions. Our technical question is therefore whether semantic retrieval adds useful connections beyond keyword search in this setting. The design must also explain those connections, disclose missing information and permit correction; finding similar text alone does not establish scientific relevance.

Plan of Approach / Design Strategy

Two trees and the knowledge boundary. Users would enter a research question, method and relevant resource needs, then choose which aspects to prioritise. A clarified proposal could be divided into experiment modules so researchers can consider reuse and replication choices separately. Keyword and semantic retrieval would search permitted records. The Similarity & Reuse Tree would group connections by question, method or resource, helping researchers

identify evidence and possible reuse. The Distinctiveness & Research Gaps Tree would describe differences from those records and questions worth investigating. Future directions belong within this second tree, avoiding a separate third output. Each branch would open a source card with the original record, the reason for the connection, its update information and any relevant missing fields. We would avoid a single similarity or novelty percentage that conceals these different relationships.

Matt Might's illustrated guide describes research as building knowledge towards a boundary and contributing beyond it (Might, n.d.). We use this as a visual metaphor: existing work supports the researcher's foundation, while differences suggest questions to investigate. Our interface can represent only the selected records, not the boundary of all human knowledge. Adding an overlooked paper could change an apparent difference. The visualisation should make this dependence clear rather than present an empty region as proof of a research gap.

Replication and researcher choice. At input, researchers could indicate whether a module aims to explore a new question, reproduce a result or identify reusable resources. Someone conducting replication could disable distinctiveness suggestions for that module while retaining references and method connections. They could accept, correct or disregard recommendations without having to justify the study to the system. This choice concerns optional advice; it does not bypass experimental safety, ethical review or access restrictions. Similarity can be valuable evidence for replication, and the same method may answer a different question.

Institutional deployment and cold start. The proposed deployment is a TU/e-managed web service on a university subdomain, using institutional hosting such as Azure if approved. SURFconext single sign-on would avoid a separate password account. Its guidance recommends minimal attribute disclosure; for returning accounts we would request a persistent pseudonymous OpenID Connect `sub` identifier (SURF, n.d.). The application would separately manage permissions for private drafts and shared records. Linking an account to Pure would require an approved mapping and user verification; we would not assume that `eduPersonPrincipalName` is a persistent cross-system key. Hosting, security review, data-protection assessment and operational ownership would be agreed with university staff before deployment. These arrangements are proposed, not secured approvals.

Cold start has two distinct forms. First, the collection must be useful before users contribute descriptions. We would seed a limited pilot with permitted Pure/portal titles, abstracts, affiliations and equipment descriptions, recording provenance and update dates. The Pure API documentation confirms that access depends on permissions and sharing configuration (TU/e, n.d.-a). Access to grants or unpublished projects cannot be assumed. We would begin with records from two departments, check coverage with their researchers and expand after correcting omissions.

Second, new users may have no publication history. Manual descriptions must therefore support the complete basic search. Users with a Pure profile could optionally review imported research information as context, then correct or reject it. Historical publications would not define their current interests. A short survey would investigate discovery needs and willingness to share; it would inform the design rather than substitute for missing research records.

For equipment omitted from abstracts, researchers or equipment managers could confirm resource tags. Integration with booking systems remains a later option: reservations do not establish actual use or suitability for another study. Automatically inferring methods and citations from bookings would introduce unsupported claims and additional personal-data processing. We would first test whether confirmed equipment descriptions meet the discovery need. The tool could direct users towards existing institutional workflows without becoming a

mandatory stage in DMP approval.

Values, participation and governance. Value Sensitive Design (VSD) connects conceptual, empirical and technical investigations (Friedman et al., 2008; VSD Lab, n.d.). Conceptually, we would examine sustainability, privacy, fairness, explicability and researcher autonomy, including people whose work appears in results but who never use the tool. Empirically, interviews and mock-up walkthroughs with researchers, students and support staff would examine useful connections, unacceptable disclosures and maintenance burdens. Technically, these findings would shape access controls, source explanations and sharing choices. Research with participants or personal data would follow TU/e's ethical-review procedures (TU/e, 2026a).

A concrete conflict concerns discovery versus confidentiality. A researcher may want others to find a reusable setup while protecting an unpublished hypothesis. Using VSD value scenarios, we would compare sharing a full description, sharing confirmed resource tags, and keeping the draft private. Participants would explain benefits and objections; the team would document its choice, affected parties and reasons, then test the revised interface with them. Our proposed default is a private draft with separately chosen sharing and contact preferences. This sacrifices some discovery coverage to protect control over unfinished work; stakeholder review would assess whether narrower sharing remains useful.

Withdrawal would remove a voluntarily shared summary from active discovery and initiate removal from derived indexes and caches. Saved views controlled by the service would hide withdrawn content and indicate that it is unavailable. This cannot guarantee deletion of external downloads or remove independently public publications. Approved retention arrangements would cover backups and necessary logs. TU/e's RDM framework defines research-data responsibilities (TU/e, 2026b). For this service, we propose a named owner coordinating corrections and withdrawals, with source and equipment owners maintaining their records under agreed responsibilities.

Evaluation and route towards impact. The pilot would compare the current portal with the proposed interface using equivalent planning tasks, balancing task order to reduce practice effects. Domain experts would judge the relevance of top results and whether explanations accurately reflect their sources. Relevance would be reported as the proportion of reviewed results judged useful; explanation errors would be claims unsupported by the linked records. We would record time to find a useful connection, missed relevant records and whether users understand the coverage limits. Results would be examined across departments and users with or without publication histories, so stronger metadata coverage does not conceal poor support for newcomers.

Users could flag an incorrect connection or outdated record. The service owner would check the source, distinguish metadata errors from retrieval or explanation errors, revise the relevant component, and repeat the affected tasks before releasing the change. A rejected suggestion would not automatically become a training label. Keeping a versioned correction record would make improvements inspectable and help determine whether errors recur.

Environmental impact requires a further chain of action: a researcher identifies a resource, its owner confirms access and suitability, the researcher changes the plan, and the resulting inputs are measured. We would compare the original and revised plans for the same scientific objective and quality requirements. Electricity and material use would be reported separately, including additional preparation, transport or cleaning where relevant and the platform's own computing costs. Uncertain estimates would remain identified as estimates. Follow-up would establish whether a suggested match actually led to reuse; clicks or accepted suggestions would not count as resource savings. Valuable replication would remain a scientific objective rather

than an avoided cost.

Comparison to Alternatives

Our Phase 1 self-assessment concept acted after an activity was defined, requiring resource measurements and reference data. Eco-log moves the intervention earlier, when finding an existing method or resource might still change the plan. It complements rather than replaces subsequent footprint assessment.

Improved portal filters and librarian-supported keyword searches are a simpler alternative. We would retain them as the baseline and fallback because they avoid some model and maintenance costs. Semantic retrieval is worth pursuing only if the pilot finds useful connections that this baseline misses, without unacceptable explanation errors or burdens. University-wide automatic profiling is another option, but requires broader permissions and risks exposing unfinished work. The limited, optional-import approach gives priority to a testable service with clear responsibilities. Its contribution is a strategy for informed research planning; effectiveness and net environmental benefit still require evaluation.

D. Responsible Innovation Project Statement

Word count: 536/ 1000

This statement examines transparency, bias and reliability in the proposed research-matching tool. The measures below address the values identified in our Value Sensitive Design approach: explicability, fairness and researchers' control over results (Friedman et al., 2008).

Transparency

Users submit a research description and receive potentially relevant previous work. They need to understand why each source was selected and be able to check the AI's explanation. We propose the following measures:

- **Source transparency:** Each source should include its title, authors, publication year and a link to the original source, with additional information where available. Researchers can then verify the sources rather than rely solely on the AI.
- **Explanation of relevance:** Each source should have a short explanation of why it was selected, identifying similarities in research questions, methods or other relevant aspects.

Bias and coverage

The tool's results may be biased by the sources available to it. For example, access limited to journal articles or English-language sources could cause researchers to miss relevant previous work. We propose the following measures to reduce this risk:

- **Database coverage:** Users should be able to see which databases the tool searches, so that they understand its limitations.
- **Search strategies:** Where possible, the search should use synonyms, similarities across disciplines and different terminology.
- **User input:** Users should be able to adjust their search query, giving them more control when the AI encounters limitations.

Reliability

AI search results can contain errors. Without checking them, users may draw incorrect conclusions: for example, the tool may present studies as similar when their methods or objectives differ. We propose the following measures:

- **Access to original:** The user should be able to inspect the papers from the results themselves. By doing this they can check whether the results are relevant to the research.
- **Limitations:** The limitations of the tool should be made clear to the user. When available information is limited, the tool should explain the uncertainty in its comparison. Additionally, finding no relevant results does not automatically mean that no previous research exists on the topic.
- **Human verification:** Users should decide whether a suggested source is relevant. The tool helps them find research; it does not decide what should happen to their plans.

Sustainable resource use

The main objective of the tool is to help the user find existing research before they spend time and resources on their project. If previous findings lead a user to revise a plan, this may prevent avoidable repetition and reduce additional material, energy, equipment and computing requirements. These savings would need to be evaluated.

However, similarity does not mean that a study is wasteful. Replication may be scientifically important, and similarities can help researchers plan it. To ensure sustainable resource use, the tool needs to provide explanations containing the similarities. Then it is up to the user how they interpret the results in terms of relevance.

Interpreting results and future work

Our design aims to make results easy to interpret through an interface that provides sources and explains their relevance. Future work could help researchers identify ways to refine their research or investigate differences from earlier work. Previous work on the same topic does not necessarily mean that a researcher should abandon a project: there may still be useful differences to explore. In such cases, the tool could also help researchers consider better use of research resources.

E. Reference List

Cohan, A., Feldman, S., Beltagy, I., Downey, D., & Weld, D. (2020). SPECTER: Document-level representation learning using citation-informed transformers. *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 2270–2282. https://doi.org/10.18653/v1/2020.acl-main.207

Friedman, B., Kahn, P. H., Jr., & Borning, A. (2008). Value sensitive design and information systems. In K. E. Himma & H. T. Tavani (Eds.), *The handbook of information and computer ethics*. Wiley. https://doi.org/10.1002/9780470281819.ch4

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SURF. (n.d.). *Attribute best practice*. https://servicedesk.surf.nl/wiki/spaces/IAM/pages/128910241/Attribute+best+practice

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Eindhoven University of Technology (TU/e). (2026b, June 9). *TU/e RDM framework policy*. https://rdm.tue.nl/docs/intro/tue-policy/

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F. User Narrative

Word count: 485/ 500

Dr. Alex sits at a desk inside the Eindhoven University of Technology chemistry lab, staring at a blank document. A new round of battery cathode tests is scheduled for next week, but Alex wants to make sure that this energy-heavy laboratory test won't just be a repeat of a study someone else has already done before. Hoping to find local equipment to share and existing data to build on, Alex opens a web browser and navigates to Eco-log.

Clicking the main button, "Explore My Research," Alex signs in instantly using his TU/e credentials via SURFconext. A clean workspace opens up, displaying a clear five-step progress tracker at the top.

Alex begins at Step 1: "Describe research". Alex types in the main research question about low-temperature cathode optimization. He adds the planned chemical methods. Then he selects TU/e as the home institution.

Moving to Step 2: "Clarify the proposal", a friendly AI consultation window pops up. It asks a quick question to clarify how the battery cells will be normalized. Alex types a short reply. The screen immediately organizes the plan into a neat checklist of three experiment modules (E1, E2 and E3).

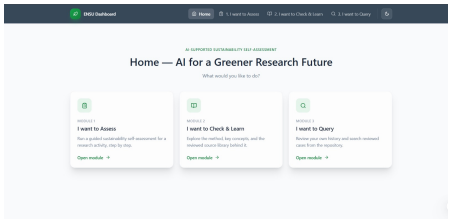
Next comes Step 3: "Review related work". Instead of a messy list of links, the dashboard organizes matching academic papers by geographic distance. Alex filters the results to show projects from the same campus and nearby regional universities. This helps Alex locate potential contacts; scientific relevance still needs checking.

This data comes to life during Step 4: "Generate two trees". Alex explores the large, interactive visual layouts. Hovering over a branch on the Similarity & Reuse Tree, a tooltip pops up showing exactly which past TU/e theses have used a matching experimental setup. Switching over to the Distinctiveness & Research Gaps Tree, Alex clicks a node labeled "No close match found in reviewed sources". This identifies a possible difference within the reviewed sources; Alex still needs to check the wider literature before claiming a research gap.

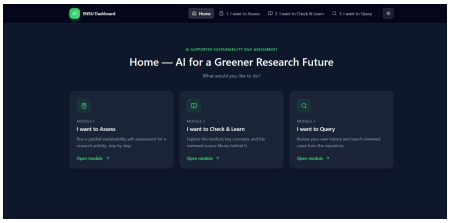
Finally, Alex reaches Step 5: "Explore recommendations". Under the "Same University" section, a card catches Alex's eye. A nearby research group right here at TU/e has an active profile listing a shared "tube furnace and coin-cell line plus a 2023 cycling dataset". A badge labeled "Avoided measurement" flags a possible reuse opportunity, not a measured saving. Alex must check whether the dataset can replace any planned measurements. The tool also lists practical next steps, advising Alex to message the lab manager to check the furnace availability.

Satisfied with the plan, Alex clicks the green "Save Assessment" button at the bottom left to securely archive his session in a private history log. Closing his laptop, Alex stands up and walks down the hallway to find the local battery team. Alex now has references, possible reuse opportunities and questions to discuss with the team. The example illustrates the proposed interaction, not verified research findings or resource savings.

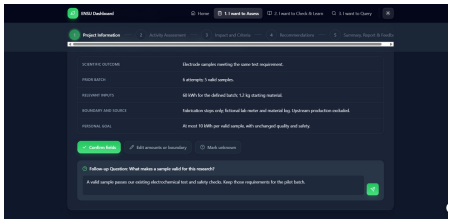
G. Visual Representation of Mock-up



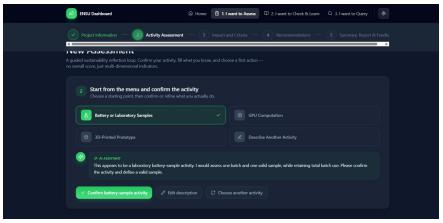
(a) Home page - Light Mode



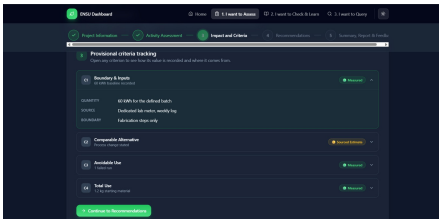
(b) Home page - Dark Mode



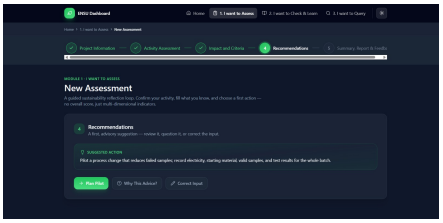
(c) I want to assess - project information



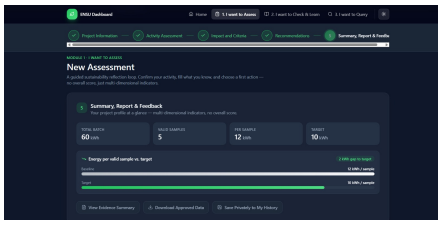
(d) I want to assess - activity assessment



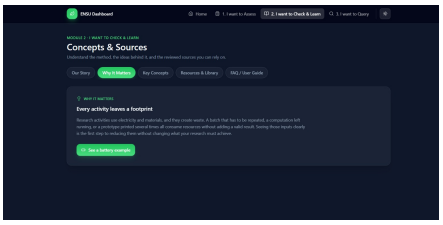
(e) I want to assess - impact and criteria



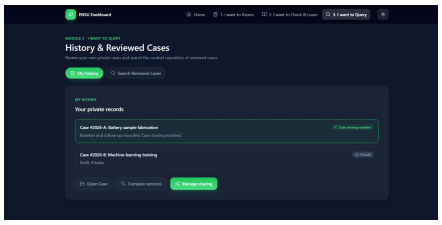
(f) I want to assess - recommendations



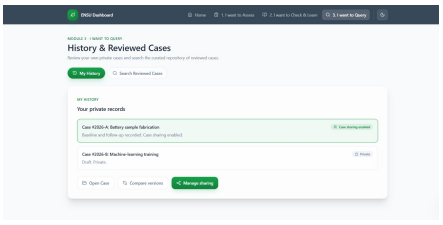
(g) I want to assess - summary, report & feedback



(h) I want to check and learn



(i) I want to query - dark mode



(j) I want to query - light mode

Figure 1: Legacy self-assessment mock-up, awaiting replacement with the five-step Eco-log interface described in Section F. These images are historical illustrations, not outputs of the current research-discovery design.

H. Documentation of Prompts, Sources, and Tools

Contributor completion notes – Gerald coordinates. These are handover requirements, not a record of completed model or API tests.

- Collect the actual prompts and refinements used for the demo, with their purpose and any model or tool version recorded by the contributor.
- Identify which sources were actually queried or imported. Distinguish documentation review from successful API access, and illustrative records from retrieved records.
- Give the real steps needed to reproduce the demo: inputs, source preparation, retrieval, output inspection and correction. Record known failures and missing dependencies.
- Document current files and editing tools. The report is maintained in `Phase2.tex`; its embedded bibliography is edited in that file. Run `scripts/compile.ps1 -Document Phase2` from the project workspace to build `output/pdf/Phase2.pdf`.
- For the written report, document AI-assisted drafting, restructuring, language editing and source checking of B/C and the editorial notes. Add the other contributors' actual tool use; do not label this work as spelling correction alone.

Editorial checklist – remove before submission

Team working notes, excluded from section word counts.

Section C – revision completed for team review

- The two trees, knowledge-boundary metaphor and module-level replication choice now form one design argument. Sources support the methods; effectiveness is not claimed as a measured result.
- Cold start now separates initial collection coverage from users without publications. Pure import is permission-dependent and optional; booking inference is deferred.
- Deployment now distinguishes hosting, authentication, application permissions and Pure identity mapping. Proposed TU/e arrangements are not represented as secured approvals.
- Peer feedback is addressed through contact preferences, a concrete confidentiality conflict, withdrawal boundaries, correction handling and resource comparisons. Dennis should develop the ethical justification in D without duplicating C's workflow description.

Section D – Dennis

- Retain the updated transparency, coverage, reliability and sustainable-resource-use sections. Confirm how the clarification chat and tree interface support human verification.
- Work through a specific conflict between discovering reusable resources and protecting unpublished ideas. Use the course framework to identify affected parties, compare options, justify a choice and define a review process.
- Specify contact preferences and the separation between using the tool and sharing a research summary.
- Explain withdrawal for live records, derived indexes, saved results and external copies; distinguish voluntarily shared summaries from independently public publications.
- Explain how incorrect links are flagged, checked, corrected and retested. A rejected suggestion is not automatically a reliable training label.
- Link researcher autonomy to replication choices and to the right to disregard suggestions. Consider users and departments with sparse records.

Remaining components – Gerald coordinates

- **A/B:** Confirm one product name across report and pitch. B has been updated to the current tool; confirm the wording against the final interface and shared example. Its maximum remains 150 words.
- **E:** Check each source against the claim it supports, as well as author names, dates, identifiers and reference style. Add sources only after consulting them. The eleven revised C references have received independent source checking; any new source added by a contributor also needs verification.
- **F:** The latest five-step narrative has been retained. Verify its illustrative records and interface labels, and add the module-level replication choice if shown in the demo; maximum 500 words.

- **G:** Replace the old screenshots with one readable A4 page matching F. Include sample input/output, two trees and a legible source card. Clearly distinguish illustrative output from actual retrieved results.
- **H:** Collect actual prompts, sources, model/tool versions where known, explored steps, limitations and reproducible hand-off instructions from the group. Replace every placeholder. Keep proposed integrations separate from completed work; disclose AI assistance accurately.
- **I:** Submit the 3–5 minute pitch separately. **J/K:** Each member submits the personal reflection and peer evaluation separately. These are not additional sections to write in this group PDF, but they are not automatically optional.
- **Final check:** Recount B/C/D/F, remove this checklist, update the draft status only when accurate, compile and inspect all pages, and verify the uploaded file and submission receipt.